# Dr. Rachel O. Cohen

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## Education

**Columbia University,** New York, NY **May 2026**
*PhD in Ecology, Evolution, and Environmental Biology*
**Columbia University,** New York, NY **May 2023**
*Master of Philosophy in Ecology, Evolution, and Environmental Biology*
**Columbia University,** New York, NY **May 2022**
*Master of Arts in Ecology, Evolution, and Environmental Biology*
**Amherst College,** Amherst, MA **May 2019**
*Bachelor of Arts in Biology* (Honors: *summa cum laude*; Distinction)

## Research Experience

**Prof. Liming Cai Lab** **University of Florida (Gainesville, FL)**
*Postdoctoral Associate* *August 2026 - Present*

- Topics: parasitic plant evolution, comparative genomics, parasitic plant population genetics, biotic interactions

**Prof. Deren Eaton Lab** **Columbia University (New York, NY)**
*PhD Candidate* *August 2020 – May 2026*

- Committee: Prof. Kevin Griffin (chair), Prof. Deren Eaton, Prof. Hilary Callahan, Prof. Andres Bendesky, Prof. Joseph Williams
- Topics: biotic interactions, pollination biology, parasitic plants, comparative genomics/transcriptomics
- Skills: genome assembly and annotation, differential gene expression quantification, DNA and RNA extractions, fluorescence microscopy, field pollinations/collections, semi-*in vivo* pollinations
- Title: "*The Genetics of Recognition: Parasitism, Reproduction, and Genome Evolution in Flowering Plants*"

**Prof. Michael Hood Lab** **Amherst College (Amherst, MA)**
*Undergraduate researcher* *January 2018 – August 2019*

- Topics: Coevolution, disease ecology, flowering plant transcriptomics

**Dr. Melissa McCormick Lab** **Smithsonian Environmental Research Center (Edgewater, MD)**
*REU Intern* *May 2018 – August 2018*

- Topics: Coevolution, symbioses in flowering plants, mycology, genetics

**Dr. Purna Chettri Research Group** **School for Field Studies Center (Paro, Bhutan)**
*Undergraduate researcher* *August 2017 – December 2017*

- Topics: Forestry, phenology, alpine ecology

## Research Interests

- Plant-plant interactions and coevolution
- Heterospecific pollination / pollination biology
- Plant parasitism
- Comparative phylogenomics, genomics and functional transcriptomics
- Community ecology

## Publications (**Bold : self**; * : undergraduate mentee)

3. Cai, L. and **R. O. Cohen**. 2026. HGTScanner: Detecting horizontal gene transfer at the fine scale using whole organelle genomes. *Methods in Ecology and Evolution.* https://doi.org/10.1111/2041-210x.70419.
2. **Cohen, R. O.**, A. Chomentowska, L. Cai, and D. A. R. Eaton. 2026. A reference genome and transcriptome of haustorial development in *Pedicularis groenlandica* reveal diverse trajectories of haustoria-associated gene evolution in parasitic plants. *Annals of Botany*. https://doi.org/10.1093/aob/mcag089.

1. **Cohen, R. O.**, A. Cisse*, J. U. Jones, J. H. Williams, and D. A. R. Eaton. 2025. Phylogeny does not predict the outcome of heterospecific pollen–pistil interactions in a species-rich alpine plant community. *American Journal of Botany*. /doi.org/10.1002/ajb2.70004.

**Manuscripts under review:**

**Cohen, R. O.,** Masi, A*., Chin, S. M., Williams, J. H., & Eaton, D. A. 2026. From recognition to neglect: Molecular and physiological responses to heterospecific pollen decay with evolutionary distance. *Evolution*. *Revisions submitted*. **Available on bioRxiv. https://doi.org/10.64898/2026.06.02.729600.**

**Manuscripts in preparation:**

## Grants and Awards

**Don Jay Melnick Award**, awarded May 2026 ("Named in honor of one of the founders of Columbia Department of Ecology, Evolution, and Environmental Biology, the Melnick Award recognizes outstanding dissertation work and other departmental activities")
**Botanical Society of America Graduate Research Award,** awarded April 2025, *$1500*
**Society for Systematic Biology Graduate Student Research Award,** awarded September 2024, *$3000*
**John Woodruff Simpson Fellowship,** ("For for the study of evolutionary biology in graduate school"), Amherst College Fellowships; received 2022-2023, 2023-2024, *$7,000* each
**Rocky Mountain Biological Laboratory Graduate Research Grant**, Rocky Mountain Biological Laboratory, April 2022, *$500*
**New York Flora Association Annual Grant**, New York Flora Association, April 2021**,** *$1500*
**Lloyd I. Rosenblum Memorial Fellowship** ("For the graduate study of Biology: Plant or Evolutionary"), Amherst College Fellowships, 2020-2021, *$10,000*
**The Oscar E. Schotté Award** (Awarded to a member of the senior class "who, in the opinion of the department, has done the best independent work in biology"), Amherst College, 2019, *$3500*
**Beckman Scholar**, Beckman Foundation, 2018, *$21,000*
**The James R. Elster Award** ("Support in the summer months for a research project to be undertaken by an undergraduate in the Department of Biology"), Amherst College, 2018, *$1200*
**The Sawyer Prize** (Awarded to the second-year student "who, in the opinion of the Biology Department at Amherst College, has shown the most promise as a student of Biology"), Amherst College 2017, *$1000*

## Teaching Experience

**Instructor Columbia Pre-College Programs**, New York, NY June 2026 - July 2026

- Taught a 3 week course titled "Environmental Studies: Designing a Sustainable Future" to talented high school students looking to take a college-level course
- Taught the course as the sole instructor and conducted all course-related teaching activities (wrote syllabus, developed lesson plans, designed activities, organized field trips).
- The course focused on biodiversity loss, climate change, and sustainable solutions to these crises.

**Columbia GSAS Teaching Scholar**, New York, NY August 2025 - Present

- Selected by the Graduate School of Arts and Sciences and E3B Department to be an instructor of record based on demonstrated teaching experience and self-designed innovative, learner-focused syllabus
- **Course title: Biological Interactions in a Changing World.** The course focuses on theory and empirical examples relating to biotic interactions and coevolution
- Taught course as the sole instructor of record in the Spring 2026 term, including preparation of all course materials (*e.g.*, lecture slides, examinations, interactive coding notebooks) as well as grading and assessment

**Teaching Fellow,** Columbia University

- Statistical Modeling for Ecology and Evolution, Fall 2022
- Fundamentals of Evolution, Fall 2021

- Introduction to Statistics for Ecology/Evolution, Fall 2020

**Teaching Assistant**, Amherst College
- Microbiology, Spring 2019
- Plant Form, Function, and Diversity, Fall 2018
- Introduction to Biology (Organism, Ecology, Evolution), Spring 2017

## Fellowships and Professional Experience

**Columbia Provost's Office**, New York, NY **September 2024 - Present**
*CURIOS Higher Education Fellow*
- CURIOS (Course-Based Undergraduate Research-Integrated Original Studies) is a provost-funded initiative aimed at implementing authentic research experiences in STEM courses
- Work with faculty across STEM disciplines (Statistics, Chemistry, Biology, Ecology/Evolution, Earth/Environmental Science, Chemical Engineering, Psychology, *etc.*) to create and implement research-based activities and lessons into undergraduate courses
- Planned and hosted workshops for faculty, created IRB for studying and reporting on the initiative, attended all meetings with core faculty team and supported needs *ad hoc*

**Columbia Graduate School of Arts and Sciences Writing Studio,** New York, NY **August 2025 - June 2026**
*Writing Consultant*
- Ran weekly group writing sessions for PhD students writing their dissertations and MA students writing their thesis
- Led hour-long one-on-one writing consultations with graduate students writing, helped with dissertation prospectuses, dissertation chapters, master's theses, postdoc fellowship applications, PhD applications
- Helped design, plan and run writing retreats for Writing Center staff and graduate students

**Columbia Center for Teaching and Learning**, New York, NY **May 2024 – June 2025**
*Senior Lead Teaching Fellow*
- Senior-level fellowship focused on pedagogy. Acted as a mentor and supervisor to 5 Lead Teaching Fellows (PhD student fellows from other departments at Columbia University)
- Advised Lead Teaching Fellows on creating and planning pedagogy-focused workshops for their respective departments and on fostering discussion about teaching within their departments
- Ran a "Learning Community", two-day workshop at the Center for Teaching and Learning for graduate student instructors focused on utilizing technology to promote active learning in the classroom

**Columbia Center for Teaching and Learning**, New York, NY **August 2022 – December 2022**
*Fellow in Academic Administration*
- Worked with the Graduate Student Programs and Services team to evaluate the reach and success of their Teaching Development Program (TDP), which provides graduate students with resources for learning about pedagogy
- Created and analyzed datasets of TDP participants in order to evaluate enrollment and completion.
- Presented findings to the Graduate Student Program team. Made written resources that are still used by the Center for Teaching and Learning to promote the TDP and its completion among graduate students.

**Columbia Center for Teaching and Learning**, New York, NY **July 2021 – May 2022**
*Lead Teaching Fellow*
- Center for Teaching and Learning representative for the Ecology, Evolution, and Environmental (E3B) Biology Department at Columbia
- Hosted two pedagogy-focused events for E3B teaching fellows. Event Fall 2021 focused on the relevance of scientific communication techniques in science education practices, and Spring 2022 focused on hosting TA office hours
- Acted as a liaison between E3B and the Center for Teaching and Learning

**The School for Field Studies**, Iquitos region, Perú **August 2019 – May 2020**
*Program Assistant*

- Oversaw daily activities and safety of undergraduate students during a semester study abroad program
- Assisted with academics, daily field lectures and exercises, as well as offered tutoring for students' independent projects, particularly with regard to statistics
- All duties were carried out on-site at the School for Field Studies Center for Amazon Studies in the Iquitos region of Perú

## Academic Outreach and Mentorship

**Mentorship & Outreach: Parasitism and pollination in alpine flowers** **May 2023 - May 2026**

- Directed projects on pollen tube growth following interspecific pollen exchange over three field seasons, oversaw fieldwork at Rocky Mountain Biological Laboratory, and molecular lab and computational work at Columbia University
- Oversaw two undergraduate projects (2023-2026), two community scientists at RMBL (one adult volunteer and one high school student, July 2025; read about that work here), and one master's student (July 2023)
- Resulted in publications with undergraduate and masters student authors (Cohen *et al.*, 2025. *American Journal of Botany* and Cohen *et al.*, 2026. Annals of Botany)
- Acted as a mentor through the Barnard College Summer Research Institute (SRI) and Barnard College Science Pathways Scholars Program ("supports talented young students from low-income or first-generation households"), May-August 2023 and May-August 2024

**Outreach: Undergraduate Outreach in Ecology and Evolution** **Oct. 2020 - May 2026**
*Committee Chair*

- Diversity and inclusion graduate student group dedicated to outreach to undergraduates at Barnard College and Columbia University with the goal of making careers in Ecology and Evolution more accessible
- Conceptualized and organized a series of professional development workshops for undergraduates: CV development, undergraduate research, and graduate school application workshops
- Work with DEI Faculty groups at Columbia and Barnard to advertise and implement our programming
- Work with Columbia introductory science courses and Columbia undergraduate clubs to share research and connect undergraduates to graduate students

**Mentorship: Ovary to Pollen Tube signaling during pollination** **March 2021 – June 2024**

- Mentored student from a New York City public high school and oversaw independent research project that investigated the role of ovary-based signaling in pollen tube growth
- Met with student weekly, taught basic lab techniques for working with plants and using microscopes

**Mentorship: Científico Latino** **August 2023 – June 2024**

- Científico Latino's Graduate Student Engagement and Community (GSEC) program "seeks to support diverse graduate students as they navigate their first year in graduate school"
- Organized monthly small-group mentorship meetings that focus on navigating graduate school

**Mentorship: RMBL Buddies Program** **Summer 2022, Summer 2023**

- Rocky Mountain Biological Laboratory summer-long mentorship program that pairs graduate students with undergraduate REU interns
- Organized weekly meetings, reviewed CV's, helped with initial graduate school search and application material development, advised on conference attendance and reviewed posters
- Mentored 3 undergraduates over 2 summers

**Mentorship: Columbia Women in Science** **Oct. 2020 – May 2023**

- Holistic mentorship program for a undergraduate students in science that are women or gender minorities
- Offered feedback on internship applications, graduate school applications, resume building, coursework, and getting involved in research in the field
- Mentored 3 different undergraduate students

**Mentorship: Amherst Women in Science**, Amherst, MA **January 2018 - May 2019**
- Acted as a mentor for underclassmen who were interested in Biology/prospective Biology majors
- Provided advice and help regarding classes, summer internships, *etc.*

**Outreach: A Better Chance Tutoring Program**, **September 2015 - May 2019**
*Community engagement leader*
- A Better Chance (ABC) is a national college preparatory program that provides room, board, and support to students from underserved school districts in high-ranking public-school districts
- Managed 10-15 college-age tutors and tutored high school students in all academic subjects, particularly Biology and Spanish, and assisted with homework, test, and college preparation

## Presentations

**Invited Presentations**
**"**The Genetics of Recognition: Parasitism, Reproduction, and Genome Evolution in Flowering Plants", September 2026, Evolution Seminar at University of Florida Biology. Gainesville, FL.
"The Evolution of Parasitism in the Flowering Plants", November 2026, Amherst College Biology Department, Amherst, MA.
"The Evolution of Parasitism in the Flowering Plants", November 2026, New York Botanical Gardens, New York, NY.

**Conference Presentations**
"Evaluating horizontal gene transfer in parasitic plants using a novel Orobanchaceae genome dataset", *Evolution* June 2025 (Oral presentation).
"Equal access opportunity: phylogeny does not predict the outcome of heterospecific pollen-pistil interactions in a species-rich alpine plant community", *International Botanical Congress* July 2024, Madrid, Spain (Oral presentation)
"Heterospecific pollen-pistil interactions illicit novel transcriptomic response relative to conspecific interactions", *International Botanical Congress* July 2024, Madrid, Spain. (Poster)
**"**A reference- and alignment-free method for assessing genotype-by-phenotype associations", *Botany* July 2021 Virtual Conference (Poster)
**"**A reference- and alignment-free method for assessing genotype-by-phenotype associations", *Evolution* June 2021 Virtual Conference (Poster)
Coevolution as the Driver of Specificity in Host-Pathogen Interactions", *Evolution* 2019, Providence, RI (Poster)
"Coevolution as the Driver of Specificity in Host-Pathogen Interactions", *Beckman Symposium* August 2019, Irvine, CA (Poster)
Coevolution as the Driver of Specificity in Host-Pathogen Interactions", *Scientista Symposium* April 2019, Boston, MA (Poster)

**Departmental Presentations**
The Genetics of Recognition: Parasitism, Reproduction, and Genome Evolution in Flowering Plants". PhD Exit Talk in Columbia E3B. April 2026. New York, NY.
"Building Inclusive Fieldwork Practices in Ecology and Evolution". Columbia E3B Seminar. September 2023, New York NY.
"Quantification of Heterospecific Pollen Tube Growth in an Alpine Plant Community", Columbia E3B Seminar. September 2022. New York, NY.

## Peer Review Activity

Journals reviewed for:

- Annals of Botany
- American Journal of Botany
- International Journal of Plant Sciences

## Workshops

Phylogenomics Software School, Evolution June 2025; Athens, GA
Phylotranscriptomics Workshop, Botany July 2022; Anchorage, AK

## Field Experience

**Colorado, USA** *June - July 2022, June – August 2023*

- Principal investigator for a project at Rocky Mountain Biological Laboratory over 3 summers, oversaw 2 field assistants (MA student & undergrad student) in summer 2023, and 2 community scientist volunteers in summer 2025
- Experimental field cross-pollinations (>3,000), plant collections, RNA and DNA sampling

**New York, USA** *April – August 2021*

- Flower collections for pollination studies in *Pedicularis canadensis* and *P. lanceolata*

**Iquitos, Perú** *August 2019 – May 2020*

- Program assistant for Peru-based program of the School for Field Studies, oversaw fieldwork of undergrads
- Forestry measurements, interviews (conducted in Spanish)

**Maritime Alps, Italy** *July 2018, July 2019*

- Disease ecology of carnations infected with fungal pathogens
- Experimental transplantation studies, flower collections, pollinator arrays, pollinator observations

**Paro, Bhutan** *August -December 2017*

- Forestry, bird surveys, flowering plant surveys, interviews

**Costa Rica (multiple locations)** *January 2017*

- Field course run by Amherst College Biology
- Plant identification, insect identification, biodiversity studies

## Relevant Coursework

***Graduate Coursework***

Fundamentals of Evolution, Statistical Modeling, Fundamentals of Ecology, Programming and Data Science for Biologists, Seminar in Race, Climate, and Environmental Justice, Field Botany (*Field intensive*), Computational Phylogenetics

***Undergraduate Coursework***

**Biology:** Adaptation and the Organism with lab, Molecules, Genes, and Cells with lab, Tropical Biology Seminar (*Field Work Intensive*), Evolution with lab (*Research Intensive*), Conservation Biology Seminar, Microbiology with lab (*Research Intensive*), Biochemical Principles
**Chemistry:** Introductory Chemistry with lab, Chemical Principles with lab, Organic Chemistry I with lab
**Mathematics/Statistics:** Multivariable calculus, Introduction to Statistics via Modeling

## Skills

Coding languages: bash, python, R.
Languages: Spanish (conversational)